

Commercial Development Vision

From portfolio prototypes to a validated surgical training product

The current projects are non-commercial educational prototypes created for my portfolio. They include a theoretical appendectomy learning module, a practical 3D appendectomy simulator, and an XR Adhesiolysis Trainer that is being adapted for immersive interaction. If developed further as a commercial product, I would use a staged process focused on clinical validity, measurable educational value, technical reliability, and gradual product scaling.

1. Clinical and user validation

The first priority would be to validate the training concept with a small group of surgeons, surgical residents, and medical students. I would use structured sessions to identify whether the content reflects real clinical workflow, whether the interaction is understandable, and which parts of the simulation provide the greatest educational value.

2. Structured feedback and learning outcomes

Feedback would be collected through short questionnaires, observation during use, and interviews. The evaluation would focus on usability, visual clarity, procedural logic, realism, confidence before and after training, and the user's ability to understand the sequence of the procedure. Where possible, simple pre- and post-training assessments could be added to measure learning rather than relying only on subjective impressions.

3. Iteration and technical improvement

Based on the results, the application would be improved in short iterations. Priorities would include clearer learning flow, better instrument interaction, more consistent feedback, improved anatomical models, and more realistic tissue behaviour. Custom 3D assets could be developed in Blender or another modelling tool to improve anatomical consistency and reduce dependence on generic assets.

4. XR development and interaction validation

The XR Adhesiolysis Trainer provides a natural next step from desktop simulation to immersive training. The focus would be on reliable instrument handling, depth perception, two-handed interaction, safe movement limits, and clear feedback during tissue manipulation. XR would be introduced only where it adds practical training value rather than as a visual feature by itself.

5. Product architecture and scaling

Instead of building one very large simulator immediately, I would develop the product as a set of focused training modules. The appendectomy project could grow from the initial access stages into a more complete traditional appendectomy pathway, while other modules could address specific skills such as adhesiolysis, suturing, or instrument handling. A modular structure would make validation, updates, and future expansion easier.

6. Commercial positioning

The primary market would be B2B education: medical universities, teaching hospitals, simulation centres, and surgical training programmes. Possible commercial models could include institutional licences, module-based licensing, or desktop/XR packages. Individual learner access could be considered later, but institutional pilots would be the preferred first step because they provide structured feedback and repeatable training environments.

7. Quality, regulatory and data strategy

Before commercial release, the intended use of each product would need to be defined clearly. A training-only application should remain separate from software intended to support diagnosis, treatment decisions, or patient-specific clinical planning. If future versions use patient data, DICOM imaging, or provide clinical decision support, the project would require a separate quality, data-protection, cybersecurity, and regulatory assessment before use in clinical care.

8. Long-term development direction

My long-term goal is to progress from medical education and XR simulation toward more advanced MedTech areas such as medical imaging, DICOM-based 3D visualisation, patient-specific surgical planning, and image-guided navigation. I would treat this as a separate development track built only after gaining stronger technical knowledge, clinical validation experience, and an understanding of the regulatory requirements for medical software.

Illustrative development roadmap

Stage	Primary objective	Example output
1. Validate	Confirm clinical relevance and usability	Pilot feedback from surgeons and learners
2. Refine	Improve interaction, anatomy and learning flow	Stable validated desktop modules
3. Expand	Add XR and a modular training library	Institutional pilot package
4. Advanced R&D;	Explore imaging, DICOM and planning	Separate prototype track with regulatory assessment

Current status: portfolio and learning projects only. This document describes a theoretical commercial development path, not a claim that the current prototypes are validated medical products.